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Atomic Units of length & energyBohr  
radius  $r$ 

$$= \frac{\epsilon_0 n^2 h^2}{ze^2 \pi m}$$

$$= \frac{n^2}{z} \times \underline{0.529 \text{ \AA}}$$

$$= \frac{n^2}{z} a_0$$

$a_0$  symbol  
of atomic  
units of  
length,  
known as  
bohr

$$1 a_0 = 0.529 \text{ \AA}$$

Bohr  
energy  $E$ 

$$E = - \frac{z^2 e^4 m}{8 n^2 \epsilon_0^2 h^2}$$

$$= - \frac{z^2}{n^2} \times 13.6 \text{ eV}$$

$$= - \frac{z^2}{2 n^2} \times \underline{(13.6 \times 2) \text{ eV}}$$

$$= - \frac{z^2}{2 n^2} E_h$$

$E_h$  symbol  
of  
atomic  
units of  
energy,  
known  
as  
hartree

$$1 E_h \approx 27 \text{ eV}$$

# Principles of Quantum Mechanics

$$\hat{H}\Psi = E\Psi$$

— This equation is known as Schrödinger Eq.

— An eigenvalue equation

$\hat{H}$  : Hamiltonian operator

$\Psi$  : Wavefunction — state of a particle  
(Atoms, molecules, electrons)

$E$  : Total energy

## Application

Structures & properties of atoms, molecules, materials at electronic level

$$\text{operator}(\text{function}) = \underbrace{\text{constant}} (\text{function})$$

→ Eigenvalue equation

Ex. 1

$$\frac{d}{dx} (e^{3x}) = \underbrace{3 \times 1}_{\text{Eigenvalue}} (e^{3x})$$

Operator

Eigenfunction

Ex. 2

$$f(x) = x^4 + 3x^2$$

$$\frac{d}{dx} (f(x))$$

$$\Rightarrow \frac{d}{dx} (x^4 + 3x^2)$$

$$\Rightarrow 4x^3 + 6x \neq f(x)$$

$$\hat{O} \psi(x) = O \psi(x)$$

↑ Wave function  
 ↑ Operator      ↑ Eigenvalue  
 [ Ex. Hamiltonian operator ]    [ Total Energy ]

General form of an eigenvalue equation

Postulate 1 : Physical observables can be obtained using quantum mechanical operators.

QM operators are hermitian & linear  
(produces real eigenvalues)

Linearity

$$\hat{O} (f_1(x) + f_2(x)) = \hat{O} f_1(x) + \hat{O} f_2(x)$$

# Learning exercise

①

Given,  $\hat{A} = \frac{d}{dx}$  &  $\hat{B} = x^2$  (multiply by)

Show that:

(a)  $\hat{A}^2 f(x) \neq [\hat{A} f(x)]^2$  } For an arbitrary  $f(x)$

(b)  $\hat{A} \hat{B} f(x) \neq \hat{B} \hat{A} f(x)$

(c) Do  $\hat{A}$  &  $\hat{B}$  commute?

(d) What comment can you make on the orders of application of operators?

② Complex numbers

③ What is the significance of imaginary no. in quantum mechanics?